

- COMPLETE ACCOUNT FLOW WITH AUDIT LOG STRUCTURE
 - I. ROLES & GLOBAL RULES
 - Roles
 - Global Rules
 - II. EMAIL POLICY (ONE EMAIL ONLY)
 - Account Created Email
 - III. AUDIT LOG STRUCTURES
 - UserAuditLog Structure
 - Metadata Fields
 - Source Values (HOW account was CREATED - immutable)
 - Context Values (WHAT action is happening - mutable)
 - FacultyAuditLog Structure
 - Metadata Fields
 - Context Values (WHAT action is happening)
 - IV. ACCOUNT FLOWS WITH COMPLETE AUDIT LOGS
 - **1** GOOGLE SSO FLOW (Faculty / Student ONLY)
 - First Google Login
 - Profile Completion
 - **2** ADMIN-CREATED FACULTY / STUDENT FLOW
 - Account Creation
 - First Login
 - Profile Completion
 - **3** ADMINISTRATOR / MCIIS STAFF FLOW
 - Account Creation
 - First Login
 - V. ACCESS CONTROL (MIDDLEWARE)
 - VI. DATABASE SCHEMA ADDITIONS
 - users table
 - VII. QUERY EXAMPLES
 - Find all Google SSO registrations
 - Find all profile completions
 - Find faculty who completed their own profile
 - Find admin-modified faculty records
 - VIII. SUMMARY OF KEY DIFFERENCES
 - IX. IMPLEMENTATION CHECKLIST
 - Phase 1: Audit Infrastructure 
 - Phase 2: Email Implementation 
 - Phase 3: First Login Tracking 

- Phase 4: Profile Completion ⌚
- Phase 5: Middleware ⌚
- Phase 6: Testing ⌚

COMPLETE ACCOUNT FLOW WITH AUDIT LOG STRUCTURE

I. ROLES & GLOBAL RULES

Roles

- Faculty
- Student
- Administrator
- MCIIS Staff

Global Rules

1. **Faculty & Student**
 - Must complete profile before system access
 - Can be created via Google SSO or Administrator
2. **Administrator & MCIIS Staff**
 - Must be pre-created by an Administrator
 - No profile completion required
3. Account creation method does NOT affect access control
4. Only ONE email is ever sent per user
5. Email is sent ONLY when an account is created by an Administrator

II. EMAIL POLICY (ONE EMAIL ONLY)

Account Created Email

 "Your account has been created"

- Sent **once**
- Sent **only** for admin-created accounts
- Purpose:
 - Account invitation
 - Login instructions
- **No other emails are sent at any point**

Google SSO users receive no emails from the system.

III. AUDIT LOG STRUCTURES

UserAuditLog Structure

Tracks **user account lifecycle** (creation, authentication, updates)

Metadata Fields

```
{
  "source": "google_sso|admin_created|seed_initialization",
  "context": "user_registration|first_login|profile_completion|role_change",
  "note": "Human-readable description",
  "google_id": "..." // Optional extra fields
}
```

Source Values (HOW account was CREATED - immutable)

Source	Constant	Meaning
google_sso	UserAuditLog::SOURCE_GOOGLE_SSO	User self-registered via Google SSO
admin_created	UserAuditLog::SOURCE_ADMIN_CREATED	Administrator created the account
seed_initialization	UserAuditLog::SOURCE_SEED_INITIALIZATION	Account created via

Source	Constant	Meaning
		database seeder

Context Values (WHAT action is happening - mutable)

Context	Constant	When Used
user_registration	UserAuditLog::CONTEXT_USER_REGISTRATION	Initial account creation
first_login	UserAuditLog::CONTEXT_FIRST_LOGIN	User's first authentication
profile_completion	UserAuditLog::CONTEXT_PROFILE_COMPLETION	User completed profile
role_change	UserAuditLog::CONTEXT_ROLE_CHANGE	User's roles were modified

FacultyAuditLog Structure

Tracks **faculty profile updates** (research interests, education, etc.)

Metadata Fields

```
{
  "context": "profile_completion|profile_update|faculty_import",
  "note": "Human-readable description"
}
```

Note: FacultyAuditLog does NOT use `source`. Use `modified_by` field to determine who made the change:

- `modified_by == target_user_id` → User updated their own profile
- `modified_by != target_user_id` → Admin updated it

Context Values (WHAT action is happening)

Context	Constant	When Used
<code>profile_completion</code>	<code>FacultyAuditLog::CONTEXT_PROFILE_COMPLETION</code>	Faculty completing profile first time
<code>profile_update</code>	<code>FacultyAuditLog::CONTEXT_PROFILE_UPDATE</code>	Subsequent profile updates
<code>faculty_import</code>	<code>FacultyAuditLog::CONTEXT_FACULTY_IMPORT</code>	Bulk import/seed operation

IV. ACCOUNT FLOWS WITH COMPLETE AUDIT LOGS

1 GOOGLE SSO FLOW (Faculty / Student ONLY)

First Google Login

Steps:

1. User logs in via Google SSO
2. System determines role:
 - Email exists in Faculty table → Faculty
 - Else → Student
3. User record is created:
 - `profile_completed = false`
 - `first_login_completed = true`
 - `email_verified_at = now()` (trusted via Google)
 - `created_by_admin = false`
4. User is redirected to profile completion page

5.  System access is blocked

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	create_user	{ "source": "google_sso", "context": "user_registration", "note": "User registered via Google SSO", "google_id": "..."} }

Controller Code:

```
UserObserver::$customMetadata = [  
    'source' => UserAuditLog::SOURCE_GOOGLE_SSO,  
    'context' => UserAuditLog::CONTEXT_USER_REGISTRATION,  
    'note' => 'User registered via Google SSO',  
    'google_id' => $googleUser->getId(),  
];  
  
$user = User::create([  
    // ... fields  
    'created_by_admin' => false,  
]);
```

Profile Completion

Steps:

1. User completes required profile
2. System updates:
 - `profile_completed = true`
 - For Faculty: Updates faculty table (research_interest, etc.)
3.  Full system access is granted
4.  No email sent

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	update_user	{ "source": "google_sso", "context": "profile_completion", "note": "Profile completed" }

Log Type	Action	Metadata
FacultyAuditLog (Faculty only)	update_faculty	{ "context": "profile_completion", "note": "Profile completed" }

Controller Code:

```
// CompleteStudentProfileController or CompleteFacultyProfileController

UserObserver::$customMetadata = [
    'source' => UserAuditLog::SOURCE_GOOGLE_SSO,
    'context' => UserAuditLog::CONTEXT_PROFILE_COMPLETION,
    'note' => 'Profile completed',
];

$user->update(['profile_completed' => true, /* ... */]);

// For Faculty only:
FacultyObserver::$customMetadata = [
    'context' => FacultyAuditLog::CONTEXT_PROFILE_COMPLETION,
    'note' => 'Faculty completed profile',
];

$faculty->update([/* research_interest, etc. */]);
```

2 ADMIN-CREATED FACULTY / STUDENT FLOW

Account Creation

Steps:

1. Administrator creates account
2. Role: Faculty or Student
3. System sets:
 - profile_completed = false
 - first_login_completed = false
 - email_verified_at = null
 - created_by_admin = true
4.  Email sent: "Your account has been created. Please login to complete profile"
5.  No system access yet

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	create_user	{ "source": "admin_created", "context": "user_registration", "note": "Account created" }

Controller Code:

```
// UserController::store()

UserObserver::$customMetadata = [
    'source' => UserAuditLog::SOURCE_ADMIN_CREATED,
    'context' => UserAuditLog::CONTEXT_USER_REGISTRATION,
    'note' => 'Account created',
];

$user = User::create([
    // ... fields
    'created_by_admin' => true,
    'profile_completed' => false,
    'first_login_completed' => false,
]);

// Send email only for Faculty/Student
if ($user->hasRole(['Faculty', 'Student'])) {
    Mail::queue(new AccountCreatedMail($user));
}
```

First Login

Steps:

1. User logs in via Google SSO
2. System updates:
 - `google_id` = ... (if Google SSO)
 - `first_login_completed` = true
 - `email_verified_at` = now()
3. User is redirected to profile completion page
4. ❌ System access is blocked
5. 📧 No email sent

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	update_user	{ "source": "admin_created", "context": "first_login", "note": "Logged in via Google SSO" }

Controller Code:

```
// FirebaseAuthController::callback() - existing user

$source = $user->created_by_admin
? UserAuditLog::SOURCE_ADMIN_CREATED
: UserAuditLog::SOURCE_GOOGLE_SSO;

UserObserver::$customMetadata = [
    'source' => $source,
    'context' => UserAuditLog::CONTEXT_FIRST_LOGIN,
    'note' => 'Logged in via Google SSO',
];

$user->update([
    'google_id' => $googleUser->getId(),
    'first_login_completed' => true,
    'email_verified_at' => now(),
]);
```

Profile Completion

Steps:

- 1. User completes profile
- 2. System updates:
 - profile_completed = true
 - For Faculty: Updates faculty table
- 3.  Full system access is granted
- 4.  No email sent

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	update_user	{ "source": "admin_created", "context": "profile_completion", "note": "Profile completed" }

Log Type	Action	Metadata
FacultyAuditLog (Faculty only)	update_faculty	{ "context": "profile_completion", "note": "Profile completed" }

Controller Code:

```

$source = $user->created_by_admin
? UserAuditLog::SOURCE_ADMIN_CREATED
: UserAuditLog::SOURCE_GOOGLE_SSO;

UserObserver::$customMetadata = [
    'source' => $source,
    'context' => UserAuditLog::CONTEXT_PROFILE_COMPLETION,
    'note' => 'Profile completed',
];

$user->update(['profile_completed' => true, /* ... */]);

// For Faculty only:
FacultyObserver::$customMetadata = [
    'context' => FacultyAuditLog::CONTEXT_PROFILE_COMPLETION,
    'note' => 'Faculty completed profile',
];

$faculty->update([/* ... */]);

```

3 ADMINISTRATOR / MCIIS STAFF FLOW

Account Creation

Steps:

1. Existing Administrator creates account
2. Role: Administrator or MCIIS Staff
3. System sets:
 - profile_completed = true (no profile completion needed)
 - first_login_completed = false
 - email_verified_at = null
 - created_by_admin = true
4.  Email sent: "Your account has been created"

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	create_user	{ "source": "admin_created", "context": "user_registration", "note": "Account created" }

Controller Code:

```
// UserController::store()

UserObserver::$customMetadata = [
    'source' => UserAuditLog::SOURCE_ADMIN_CREATED,
    'context' => UserAuditLog::CONTEXT_USER_REGISTRATION,
    'note' => 'Administrator created admin/staff account',
];

$user = User::create([
    // ... fields
    'created_by_admin' => true,
    'profile_completed' => true, // No profile completion needed
    'first_login_completed' => false,
]);

// Always send email for Admin/Staff
Mail::queue(new AccountCreatedMail($user));
```

First Login

Steps:

1. User logs in via Google SSO
2. System updates:
 - `google_id = ...` (if Google SSO)
 - `first_login_completed = true`
 - `email_verified_at = now()`
3.  Immediate system access (no profile completion needed)
4.  No email sent

Audit Logs:

Log Type	Action	Metadata
UserAuditLog	update_user	{ "source": "admin_created", "context": "first_login", "note": "Logged in via Google SSO" }

Controller Code:

```
// FirebaseAuthController::callback() - existing user

$source = $user->created_by_admin
? UserAuditLog::SOURCE_ADMIN_CREATED
: UserAuditLog::SOURCE_GOOGLE_SSO;

UserObserver::$customMetadata = [
    'source' => $source,
    'context' => UserAuditLog::CONTEXT_FIRST_LOGIN,
    'note' => 'Admin/Staff logged in via Google SSO',
];

$user->update([
    'google_id' => $googleUser->getId(),
    'first_login_completed' => true,
    'email_verified_at' => now(),
]);

// Redirect to browse (no profile completion)
return redirect()->route('browse');
```

V. ACCESS CONTROL (MIDDLEWARE)

```
// EnsureProfileCompleted Middleware

if (in_array($user->role, ['Faculty', 'Student']) && !$user->profile_completed) {
    // Redirect to profile completion page
    return redirect()->route('profile.complete');
}

// Creation source does not matter - both Google SSO and admin-created follow same rule
```

VI. DATABASE SCHEMA ADDITIONS

users table

```
$table->boolean('created_by_admin')->default(false)
    ->comment('Whether account was created by administrator (true) or via Google SSO
(false)');
```

Usage:

- Determines **source** in UserAuditLog metadata
 - Query: `User::where('created_by_admin', true)->get()` (all admin-created accounts)
-

VII. QUERY EXAMPLES

Find all Google SSO registrations

```
SELECT * FROM user_audit_logs
WHERE action_type = 'create_user'
      AND JSON_EXTRACT(metadata, '$.source') = 'google_sso';
```

Find all profile completions

```
SELECT * FROM user_audit_logs
WHERE action_type = 'update_user'
      AND JSON_EXTRACT(metadata, '$.context') = 'profile_completion';
```

Find faculty who completed their own profile

```
SELECT * FROM faculty_audit_logs
WHERE action_type = 'update_faculty'
      AND JSON_EXTRACT(metadata, '$.context') = 'profile_completion'
      AND modified_by = target_user_id;
```

Find admin-modified faculty records

```
SELECT * FROM faculty_audit_logs
WHERE modified_by != target_user_id;
```

VIII. SUMMARY OF KEY DIFFERENCES

Aspect	UserAuditLog	FacultyAuditLog
Purpose	User account lifecycle	Faculty profile updates
Source field	✔ Yes (how account created)	✗ No (use modified_by)
Context field	✔ Yes (lifecycle phase)	✔ Yes (action type)
Create action	✔ Yes (account creation)	✗ Rarely (only during import)
Update action	✔ Yes (login, profile)	✔ Yes (profile changes)
Who did it	Check modified_by	Check modified_by

IX. IMPLEMENTATION CHECKLIST

Phase 1: Audit Infrastructure

-  UserAuditLog constants (SOURCE_, CONTEXT_)
-  FacultyAuditLog constants (CONTEXT_* only)
-  UserObserver structureMetadata() method
-  FacultyObserver structureMetadata() method
-  Migration: created_by_admin column

Phase 2: Email Implementation

- ☐ Create AccountCreatedMail class
- ☐ Update UserController::store() to send emails
- ☐ Set metadata for admin account creation

Phase 3: First Login Tracking

- ☐ GoogleAuthController already handles this (verify)
- ☐ AuthenticatedSessionController for non-Google logins

Phase 4: Profile Completion

- ☐ CompleteStudentProfileController metadata
- ☐ CompleteFacultyProfileController metadata

Phase 5: Middleware

- ☐ EnsureProfileCompleted - verify logic

Phase 6: Testing

- ☐ Test all three flows
- ☐ Verify audit logs
- ☐ Verify email sending